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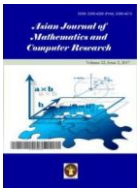
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STUDY ON ALLOCATION OF PRODUCTION RESOURCES FOR AN OPTIMAL PROFIT MAKING

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This work was carried out in collaboration among all authors. All authors read and approved the final manuscript.

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Abstract

In this study, the limited resources of a production company are maximally allocated for the production of cake of different sizes for optimal profit making. The datasets used were collected on the profit made from four different sizes of the cake and on the available production resources of the company. The profits made from the various cake sizes form the objective function coefficients; the measurements of the various resources used for the production of the different sizes of the cake form the coefficients in the constraint equations while the total available resource quantities form the right hand sides of the constraint equations. The resultant linear programming problem was solved and the result showed the various quantities of the cake sizes that should be produced so as to make an optimal profit. Sensitivity analysis was carried out to determine the robustness of the formulated model.

Keywords: Linear programming; objective function; sensitivity analysis; optimal solution.

1 Introduction

Every profit oriented organization aims at maximizing profit with their available limited resources. Adequate production planning and execution are therefore required in order to achieve this purpose and also maintain a lasting existence of such organization [1,2 and 3]. Also, every production company has a number of product lines which consists of the product-mix of such a company [4]. Product-mix determination enables company to focus its production on relevant products that enhance profit making [5,6,7,4,8 and 9]. Nonetheless, a company may be tempted to add more product lines in their quest to have more customers; thereby producing such products that appeal only to a fraction of its customer base.

Since different products require different amounts of production resources, different costs and profits at different stages of production, there is need for a mathematical technique that can determine the product mix that will maximize the total profit [4,10,11,9 and 12]. This mathematical technique is called "Linear Programming (LP)"

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[13, 14, 15, 16 and 8]. This technique uses model to represents a real life situation or system for competent decision making and analysis [14, 17, 18, 15, 19, 6 and 10].

Rama et al. [3] used LP model to schedule drivers in a transport co-operation so as to determine the minimum number of drivers needed for each shift in a day in order to reduce the cost of reserving drivers. [4 and 9] considered linear programming model as a quantitative decision making tool needed for proper optimization of product mix [7] used big M and dual simplex methods as alternative methods for finding solutions to linear programming that can reduce the number of iteration and also save computer time [13] used the simplex algorithm of linear programming for the optimal allocation of raw materials for the production of different sizes of bread in a bakery industry. [20,21,4and10] used simplex method of linear programming to determine the maximum profits of the concerned companies.

In every optimization problem, due to the limited nature of resources, it is always of interest not only to obtain an optimal solution of the problem but also to find how stable that optimal solution would be, assuming there changes either the objective function's cost coefficient or the right hand side (RHS) resource constraints. In this study, sensitivity analysis was carried out so as to know how an optimal solution of the mathematical programming problem changes due to changes in the input data. This work also considered sensitivity analysis on the cost coefficients of the objective function and the resource constraints of the problem so as to ascertain the robustness of the model.

2 Motivation of the Study

Since this study aims at determining how the limited resources (raw materials) of the bakery company could be allocated to the product-mix in order to maximize profit; the researcher felt that due to the changing trends in the business world, this allocation should not be done arbitrarily since there is no guarantee of achieving the intended purpose. The company, according to its own records, has never used any scientific procedure for her production decision making and as a result, could not boast of making their desired profit over the past years. This work therefore provides the right production template for the company

3 Materials and Methodology

3.1 Material

The data used in this study presented in Tables 1 and 2 were collected from the concerned bakery company's day to day production record on each of the four sizes of cake. The company's target is to maximize the daily profit using the available resources. Data were therefore collected on:

1) The raw materials available.
2) The profit made from the sale of each size of the cake.
3) The different measurements of the resources needed for each product mix.

The raw materials used and their respective quantities are as shown in the Table 1.

3.2 Methodology

3.2.1 Linear programming model

In order to help the company have a structured process with documented steps and measurable results in allocating their raw materials for optimal production that will enhance the profit making of the business, the linear programming model given below has been considered the most efficient.

$$\text{Maximize } f(x) = c_1x_1 + c_2x_2 + c_3x_3 + \ldots + c_nx_n = \sum_{i=1}^{n} c_ix_i \quad (1)$$

$$\text{Subject to: } g_i(x) = a_{i1}x_1 + a_{i2}x_2 + a_{i3}x_3 + \ldots + a_{in}x_n = \sum_{j=1}^{m} a_{ij}x_j \leq b_i \quad (2)$$

$$x \geq 0 \quad (3)$$

where f (x) is the objective function to be optimized..

$x = (x_1, x_2, \ldots, x_n)^T$ represents the feasible solution that maximizes the profit.

$x \geq 0$ restricts the values of x to nonnegative real numbers.

$g_i(x)$ is the constraint function of x

b_i is the resource constraint

Equations (1), (2) and (3) can be expressed in matrix form as:

$$\begin{aligned} &\text{Maximize } Z = C^T X \\ &\text{Subject to } AX \leq b \\ &X \geq 0 \end{aligned} \tag{4}$$

where $X \epsilon R^n$ is a column vector of variables to be determined

$C \epsilon R^n$ and $b \epsilon R^m$ are column vectors of cost coefficient and right hand side parameters, respectively. $A \epsilon R^{mxn}$ is a constraint coefficient matrix, and the superscript T is the transpose operator.

The standard form of the model in (4) is given in (5) as

$$\left.\begin{aligned} &\text{Max } Z = C_1x_1 + C_2x_2 + C_3x_3 + \ldots + C_nx_n \\ &\text{S.t} \\ &a_{11}x_1 + a_{12}x_2 + \ldots + a_{1n}x_n + x_{n+1} = b_1 \\ &a_{21}x_1 + a_{22}x_2 + \ldots + a_{2n}x_n + x_{n+2} = b_2 \\ &\vdots \\ &a_{m1}x_1 + a_{m2}x_2 + \ldots + a_{mn}x_n + x_{n+m} = b_m \end{aligned}\right\} \tag{5}$$

For this study, C_i, $i = 1, 2, \ldots, 4$ is the per unit profit of each size of the cake ; a_{ij} is the quantity of each resource j needed for the production of each cake size i and b_j, $j = 1,2,\ldots,6$ is the total quantity of each resource, j available.

The decision variables, x_i, $i = 1, 2, 3, 4$ represent the four different sizes of the cake produced by the bakery company:

3.2.2 Simplex algorithm

The steps involved in simplex algorithm of linear programming are as follows:

i. Formulate the linear programming problem (model) to be optimized using the appropriate decision variables.
ii. Transform the model constraints into equations by adding either slack-variables, or surplus variables. The slack variable is used when the constraint has 'less than or equal to (≤) sign' while the surplus variable is used when the constraint has 'greater than or equal to (≥) sign'
iii. Put the problem is arrayed in tableau with columns and rows.
iv. Determine the pivot column (entering variable) by selecting the column with the highest negative value along the *Cj-Zj* row.
v. Determine the pivot row (leaving variable) by dividing the solution column values by the pivot-column values and selecting the row with the highest minimum non-negative quotient.

vi. Change the value at point of intersection of pivot column and row to one by dividing the entire row by the same value.
vii. Use Echelon row reduction technique to change every other value in the pivot column to zero
viii. Obtain the new *Zj* and *Cj-Zj* rows.
ix. Determine whether the new solution is optimal by checking the *Cj – Zj* row. If all *Cj – Zj* row values are zeros or positive, the solution is optimal otherwise return to step (vi) and repeat the process.

3.2.3 Sensitivity analysis

The optimal solution of the linear programming problem (1), (2), (3) or (4) depends upon the parameters C_j ,a_{ij} and b_i of the problem. If the optimal value of the objective function is relatively sensitive to changes in certain parameters, then the estimation of such parameters and the selection of a solution which does well for most of their likely values should be done carefully. It is also important to know the range of values for which the solution is both optimal and feasible. Such investigation as described above is called sensitivity analysis. The changes in linear programming problems studied by sensitivity analysis include:

(1) The objective function coefficients
(2) Changes in the right hand side constants

In order to obtain the range of values of the objective function cost coefficients for which the solution remains optimal, we use the formular below:

$$Max_j \left\{ \frac{-\left(Z_j - C_j\right)/t_{rj} \geq 0}{t_{rj}} \right\} \leq \Delta C_j \leq Min \left\{ \frac{-\left(Z_j - C_j\right)/t_{rj} \leq 0}{t_{rj}} \right\}$$

where Z_j - C_j is the value in the *Z*-equation of the final iteration tableau which contains the optimal solution and t_{rj} is the constraint coefficient of the basic variable whose range of the objective function coefficient is being sought for.

Also, suppose b_k changes from

$$\hat{b}_k = b_k + \Delta b_k \ell_k$$

where $\hat{b}_k = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_k + \Delta b_k \\ \vdots \\ b_m \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_k \\ \vdots \\ b_m \end{bmatrix} + \Delta b_k \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix}$

the old optimal solution must remain optimal since optimality criterion does not depend on *b*. However, the old solution may no longer be feasible. Hence the need to compute the new

$$\hat{X}_B = B^{-1}b + \Delta b_k B^{-1}\ell_k;$$
$$\hat{X}_B = X_B + \Delta b_k B^{-1}\ell_k$$

where $X_B = B^{-1}b$

$B^{-1}\ell_k$ is the kth column of the matrix B^{-1}. This column appears in the final simplex tableau in the column which holds the kth column of the identity matrix of the initial simplex tableau. So, to have the solution remain feasible, Δb_k must be chosen such that

$$X_B + \Delta b_k B^{-1}\ell_k \geq 0$$

4 Data Analysis

In order to obtain the best product mix that will lead to maximization of the company's profit and also determine the range of changes in the cost coefficients of the objective function and the resource constraint, simplex method procedures outlined in Section 2 were applied to the data presented in Tables 1 and 2.

Table 1. Raw materials and their quantities in grams and kilograms

Ingredient/Measurement	Size 1 (X_1)	Size 2 (X_2)	Size 3 (X_3)	Size 4 (X_4)	Total
Flour(kg)	0.7	I.0	1.3	1.5	35
Sugar(g)	0.2	0.3	0.5	0.7	100
Egg(kg)	0.2	0.5	1.0	1.2	50
Butter(kg)	0.2	0.4	0.6	0.9	67
Baking powder(g)	0.1	0.2	0.4	O.7	10
Water(kg)	0.3	0.7	1.0	1.3	300

Table 2. Cake sizes with their per unit profit

Product	per unit Profit (₦)
X_1	500
X_2	700
X_3	900
X_4	1200

4.1 Linear Programming Model formulation

The model consists of the objective function and the constraints as shown in (6)

$$\begin{aligned} \text{Max } Z &= 500X_1 + 700X_2 + 900X_3 + 1200X_4 \\ \text{St} \quad & 0.7X_1 + 1.0X_2 + 1.3X_3 + 1.5X_4 \leq 35 \\ & 0.2X_1 + 0.3X_2 + 0.5X_3 + 0.7X_4 \leq 100 \\ & 0.2X_1 + 0.5X_2 + X_3 + 1.2X_4 \leq 50 \\ & 0.2X_1 + 0.4X_2 + 0.6X_3 + 0.9X_4 \leq 67 \\ & 0.1X_1 + 0.2X_2 + 0.4X_3 + 0.7X_4 \leq 10 \\ & 0.3X_1 + 0.7X_2 + X_3 + 1.3X_4 \leq 300 \\ & X_i \geq 0; i = 1, 2, 3, \text{ and } 4 \end{aligned} \tag{6}$$

The problem in (6) can be transformed by introducing the slack variables S_i and changing the inequality to equality as follows:

$$\begin{aligned} & Z - 500X_1 - 700X_2 - 900X_3 - 1200X_4 = 0 \\ \text{St} \quad & 0.7X_1 + 1.0X_2 + 1.3X_3 + 1.5X_4 + S_1 = 35 \\ & 0.2X_1 + 0.3X_2 + 0.5X_3 + 0.7X_4 + S_2 = 100 \\ & 0.2X_1 + 0.5X_2 + X_3 + 1.2X_4 + S_3 = 50 \\ & 0.2X_1 + 0.4X_2 + 0.6X_3 + 0.9X_4 + S_4 = 67 \\ & 0.1X_1 + 0.2X_2 + 0.4X_3 + 0.7X_4 + S_5 = 10 \\ & 0.3X_1 + 0.7X_2 + X_3 + 1.3X_4 + S_6 = 300 \\ & X_i \geq 0; i = 1, 2, 3, 4; \; S_i \geq 0; i = 1, 2, \ldots, 6 \end{aligned} \tag{7}$$

At this point, the problem in (7) is presented in tableau form and simplex algorithm applied with series of iterations until an optimal solution is obtained. The process proceeds as follows:

Table 3. Initial tableau of the simplex algorithm

	Z	**X_1**	**X_2**	**X_3**	**X_4**	**S_1**	**S_2**	**S_3**	**S_4**	**S_5**	**S_6**	**b_i**
Z	1	-500	-700	-900	-1200	0	0	0	0	0	0	0
S_1	0	0.7	1.0	1.3	1.5	1	0	0	0	0	0	35
S_2	0	0.2	0.3	0.5	0.7	0	1	0	0	0	0	100
S_3	0	0.2	0.5	1.0	1.2	0	0	1	0	0	0	50
S_4	0	0.2	0.4	0.6	0.9	0	0	0	1	0	0	67
S_5	0	0.1	0.2	0.4	0.7	0	0	0	0	1	0	10
S_6	0	0.3	0.7	1.0	1.3	0	0	0	0	0	1	300

From Table 3, it is discover that there is no optimality since the coefficients of the non-basic variables, X_1, X_2, X_3 and X_4 in the *Z*-equation are all non positive. The next step is to look for the entering variable which is the one with the highest negative coefficient. X_4 is the entering variable. At this juncture, decision is made for the variable that should leave the basic column for this entering variable and it is the variable that has the least coefficient after dividing the values of the variables in the solution column by the corresponding values under the entering variables. Concerning the problem above, X_4 is the entering variable while S_5 is the leaving variable. Another table is now created in order to achieve the numbers *v, vi* and *vii* of the simplex algorithm.

Table 4. First iteration

	Z	**X_1**	**X_2**	**X_3**	**X_4**	**S_1**	**S_2**	**S_3**	**S_4**	**S_5**	**S_6**	**b_i**
Z	1	-328.57	-357.14	-214.29	0	0	0	0	0	1714.29	0	17142.86
S_1	0	0.49	0.57	0.44	0	1	0	0	0	-2.14	0	13.57
S_2	0	0.1	0.1	0.1	0	0	1	0	0	-1.0	0	90.0
S_3	0	0.03	0.16	0.31	0	0	0	1	0	-1.71	0	32.86
S_4	0	0.07	0.14	0.09	0	0	0	0	1	-1.29	0	54.14
X_4	0	0.14	0.29	0.57	1	0	0	0	0	1.43	0	14.29
S_6	0	0.11	0.33	0.26	0	0	0	0	0	-1.86	1	281.43

Considering Table 4, it is easily discovered that optimality has not yet been achieved since the coefficients of the non-basic variables, X_2 and X_3 are all negative. X_2 the entering variable since it has a higher negative coefficient than X_3 while S_1 is the leaving variable since it has a smaller quotient. Procedures in numbers *v, vi* and *vii* are also carried out, Table 5.

Table 5. Second iteration

	Z	**X_1**	**X_2**	**X_3**	**X_4**	**S_1**	**S_2**	**S_3**	**S_4**	**S_5**	**S_6**	**b_i**
Z	1	-25	0	62.5	0	625	0	0	0	375	0	25625
X_2	0	0.85	1	0.78	0	1.75	0	0	0	-3.75	0	23.75
S_2	0	0.02	0	0.02	0	-0.18	1	0	0	-0.63	0	87.63
S_3	0	-0.11	0	0.19	0	-0.28	0	1	0	-1.13	0	29.13
S_4	0	-0.05	0	-0.03	0	-0.25	0	0	1	-0.75	0	50.75
X_4	0	0.10	0	0.35	1	-0.5	0	0	0	2.5	0	7.5
S_6	0	-0.17	0	0	0	-0.58	0	0	0	-0.63	1	273.63

From Table 5, it is also seen that optimality has not yet been achieved since the coefficients of the non-basic variables, X_1 is negative. Therefore, X_1 is the entering variable while X_2, having a smaller quotient is the leaving variable. This decision leads to the table 6.

Table 6. Final iteration

	Z	**X_1**	**X_2**	**X_3**	**X_4**	**S_1**	**S_2**	**S_3**	**S_4**	**S_5**	**S_6**	**b_i**
Z	1	0	29.41	85.29	0	676.47	0	0	0	264.71	0	26323.53
X_1	0	1	1.18	0.91	0	2.06	0	0	0	-4.41	0	27.94
S_2	0	0	-0.02	0.01	0	-0.21	I	0	0	-0.56	0	87.21
S_3	0	0	0.12	0.29	0	-0.06	0	1	0	-1.59	0	32.06

	Z	X_1	X_2	X_3	X_4	S_1	S_2	S_3	S_4	S_5	S_6	b_i
S_4	0	0	0.06	0.02	0	-0.15	0	0	1	-0.97	0	52.15
X_4	0	0	0.12	0.44	1	-0.29	0	0	0	2.06	0	10.29
S_6	0	0	0.19	0.15	0	-0.24	0	0	0	-1.35	0	278.24

Table (6) shows that optimality has been achieved since all the cost coefficients of the *Z*-equation are positive. Hence, $Z = 26323.53$; $X_1 = 27.94$; $X_4 = 10.29$

4.2 Determination of sensitivity analysis on profits

Here, the range of change on the profit (ΔC_i, i =1, 2, 3 and 4) made on each cake size that guarantees the stability of the optimal solution obtained in (6) is each obtained. From Table 6, it can easily be seen that the profit coefficients of X_2 and X_3which are non-basic variables can be increased by 29.41and 85.29 respectively without changing the optimal solution since all the *Z*-values are positive. The ranges of changes in C_2 and C_3 within which the solution remains optimal as can be seen in (6) are as given below:

$$-\infty \le \Delta C_2 \le 29.41$$

Adding the original profit coefficient (700) in (6) to both sides, we have that

$$-\infty \le \Delta C_2 \le 729.41$$

For C_3, $-\infty \le \Delta C_3 \le 85.29$

Adding the original profit coefficient (900) in (6) to both sides, we have that

$$-\infty \le \Delta C_3 \le 985.29$$

We now try to obtain the ranges for which changes in the profit coefficients of X_1 and X_4 (which are basic variables) will still allow the solution to remain optimal using:

$$Max_j\left\{\frac{-\left(Z_j - C_j\right)/t_{rj} \ge 0}{t_{rj}}\right\} \le \Delta C_j \le Min\left\{\frac{-\left(Z_j - C_j\right)/t_{rj} \le 0}{t_{rj}}\right\}$$

For X_1, $Max_j\left\{\frac{-29.41}{1.18}, \frac{-85.29}{0.91}, \frac{-676.47}{2.06}\right\} \le \Delta C_1 \le Min_j\left\{\frac{-264.71}{-4.41}\right\}$

$= Max_j\{-24.9, -93.7, -328.4\} \le \Delta C_1 \le Min_j\{60\}$

$= -25 \le \Delta C_1 \le 60$

Since the original profit coefficient associated to X_1 is 500 as can be seen also in (6), we add 500 to both sides of the inequality to have that

$$475 \le \Delta C_1 \le 560$$

For X_4, $Max_j\left\{\frac{-29.41}{0.12}, \frac{-85.29}{0.44}, \frac{-264.71}{2.06}\right\} \le \Delta C_4 \le Min_j\left\{\frac{-676.47}{-0.29}\right\}$

$Max_j\{-245, -194, -129\} \le \Delta C_4 \le Min_j\{2333\}$

$-129 \le \Delta C_4 \le 2333$

Since the original profit coefficient associated to X_4 is 1200 as shown in (6), we add 1200 to both sides of the inequality to have that

$$1071 \leq \Delta C_4 \leq 3533$$

The summary of the results is presented in Table 7.

Table 7. Summary of the sensitivity Analysis on Profit

Profit	Minimum	maximum
C_1	475	560
C_2	$-\infty$	729.41
C_3	$-\infty$	985.41
C_4	1071	3533

4.3 Determination of sensitivity analysis on RHS resources

In order to obtain the minimum and maximum values of the RHS resource constraints for which the solutions remains feasible, the matrix in the final table of the optimal solution that corresponds to the identity matrix of the initial table is multiplied by the column of the resource vectors ($B^{-1}b_i$) as follows:

$$\begin{bmatrix} 2.06 & 0 & 0 & 0 & -4.41 & 0 \\ -0.21 & 1 & 0 & 0 & -0.56 & 0 \\ -0.06 & 0 & 1 & 0 & -1.59 & 0 \\ -0.15 & 0 & 0 & 1 & -0.97 & 0 \\ -0.29 & 0 & 0 & 0 & 2.06 & 0 \\ -0.24 & 0 & 0 & 0 & -1.35 & 1 \end{bmatrix} X \begin{bmatrix} 35 \\ 100 \\ 50 \\ 67 \\ 10 \\ 300 \end{bmatrix} \geq 0$$

For all the RHS resource constraints $b^{'s}$, when each is replaced with its corresponding b_i , i=1, 2, 3, …, 6 and the two matrices are multiplied with the inequality ≥ 0 attached, the following results are obtained:

$21.43 \leq b_1 \leq 1{,}194$, $b_2 \geq 12.95$, $b_3 \geq 18$, $b_4 \geq 14.95$, $b_5 \geq 5$ and $b_6 \geq 21.9$

The summary of the results is presented in Table 8.

Table 8. Summary of the sensitivity analysis on the resources

Resources	Minimum value	Maximum value
b_1	21.43	1,194
b_2	12.95	$+\infty$
b_3	18	$+\infty$
b_4	14.95	$+\infty$
b_5	5	$+\infty$
b_6	21.9	$+\infty$

5 Discussion of Results

The result in table 6 shows the values of the decision variables as X_1 = 27.94; X_2= 0; X_3 = 0 and X_4 = 10.29 with the optimal profit of Z = ₦26,323.53. This implies that the second and third sizes of the company's cake do not contribute to the profit making of the company while the quantities of the first and the fourth sizes needed by the company to make a profit of ₦26,323.53 are 27.94 and 10.29with ₦13,970.98 and ₦12,352.44 as their respective profit contributions. The sensitivity analysis results summarized in Table 7 show that the profit to be targeted by the company on the smallest size cake should not be less than ₦475 neither should it be more than ₦560 if the optimal solution must be maintained. Also, for the fourth cake size, the profit should neither be less than ₦1,071 nor be more than ₦3533. The same table also shows that reducing the amounts of profits made on

the second and the third sizes of the company's cake even to the barest minimum cannot attract any profitable patronage from the customers. Table 8 shows that the quantity of flour needed by the company should neither be less than 21.43Kg nor be more than 1,194Kg in order to make their desired profit. Also, other resources like sugar, egg, butter, baking powder and water should not be less than 12.95Kg, 18Kg, 14.95Kg, 5g and 21.9 litres respectively if the desired optimal profit of the company should not be violated.

6 Conclusion

This work aims at modeling the production activities of a bakery company that produces four different sizes of cake so as to know the optimal profit that the company is likely to make and the quantity of each cake size to be produced with their individual profit contributions. The data collected from the company was analysed using simplex algorithm and the results clearly specified the ranges of the profits to be made on each of the products in order to ensure optimal profit making. The quantities of the resources to be made available during production were also specified. It is therefore concluded that the results of this work provided a production template for the bakery company in question.

Competing Interests

Authors have declared that no competing interests exist.

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